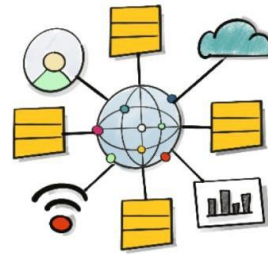
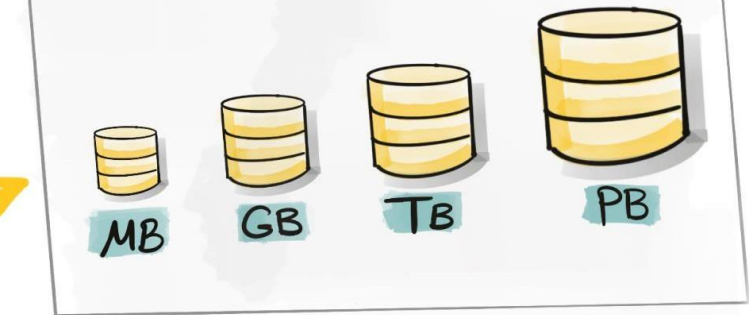


VELOCITY

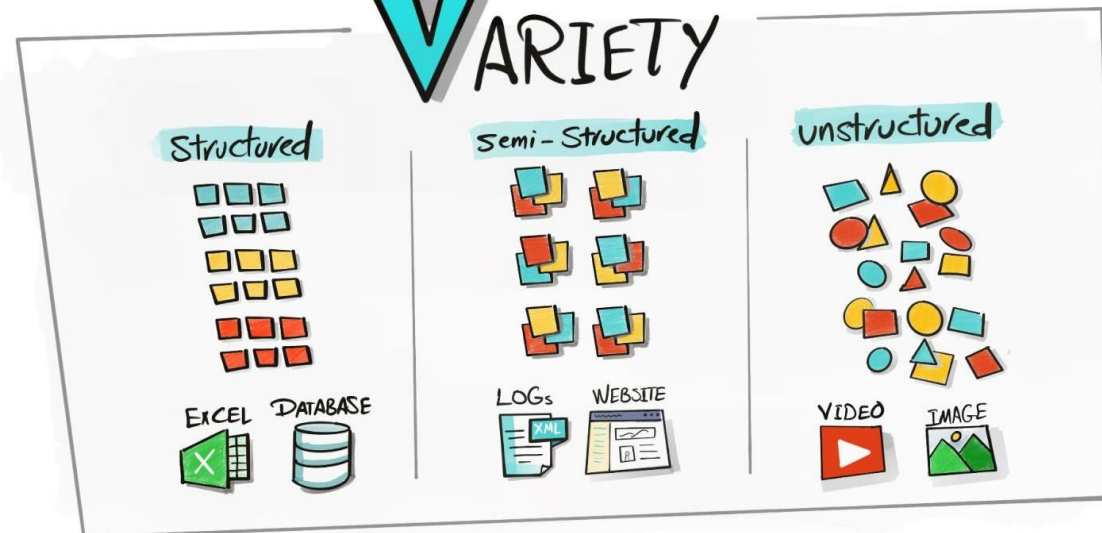


BIG DATA

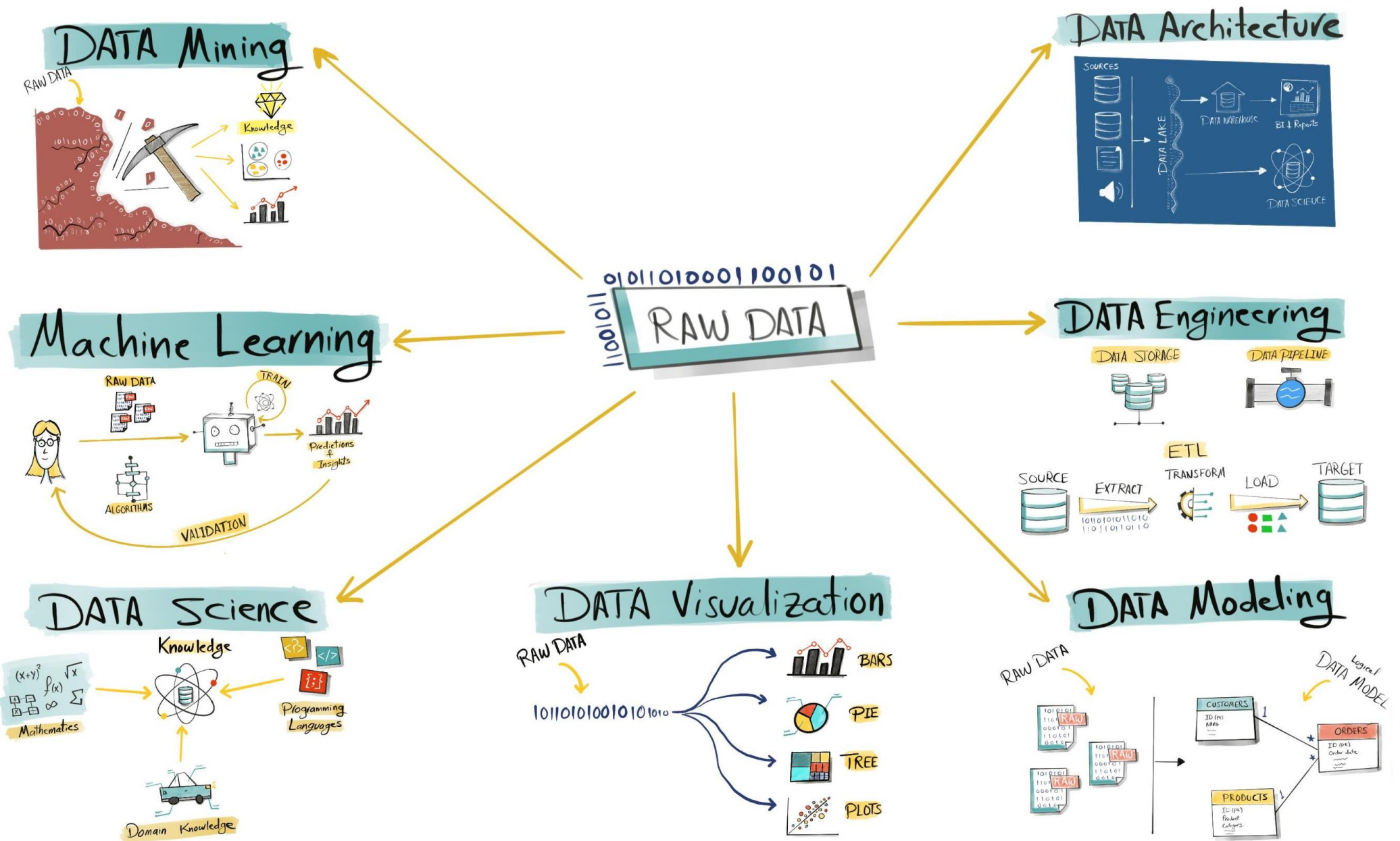
VOLUME

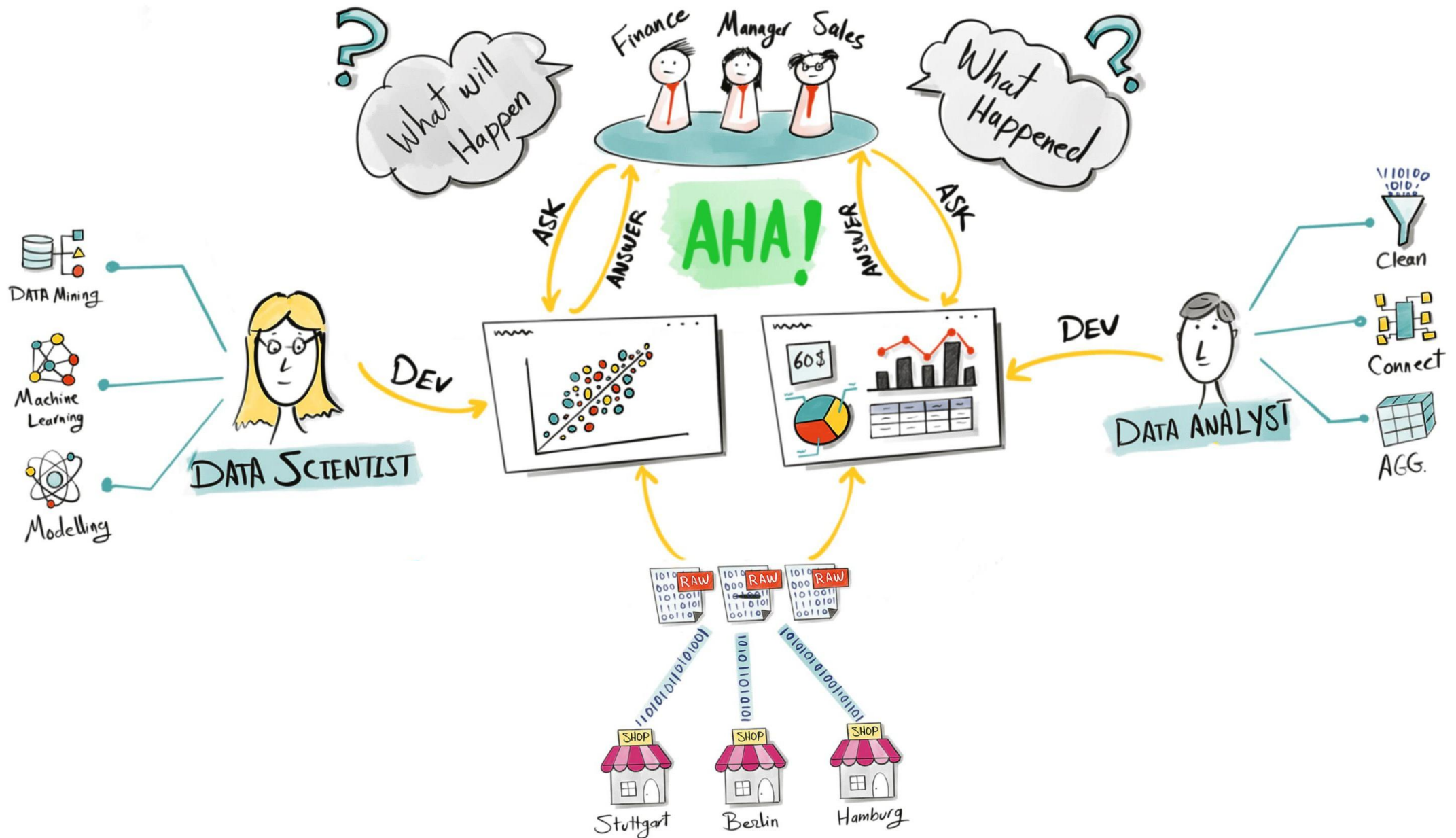


VARIETY

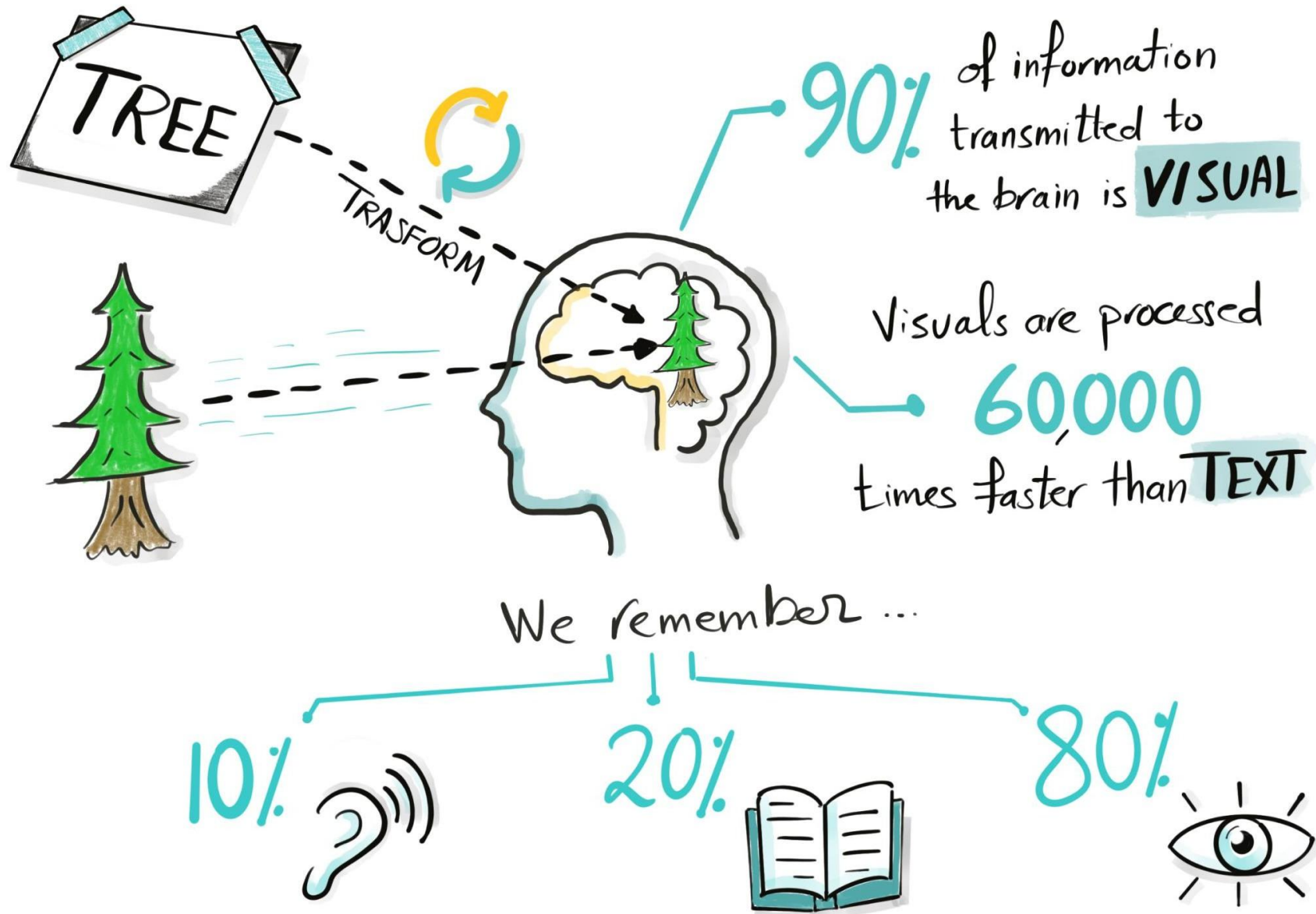


DATA BUZZWORDS





POWER OF VISUAL



BENEFITS OF DATA VIZ



Benifits of Tableau vs Excel



Automation ✓



Security ✓



BIG DATA ✓



Interactive ✓



DEVELOPEMENT PROCESS

①
CONNECT



②
PREPARE

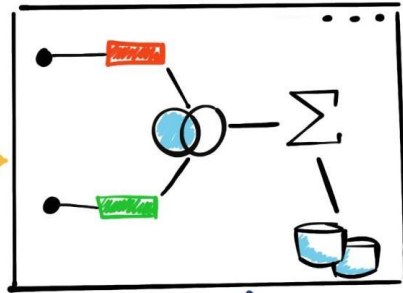


Tableau PREP

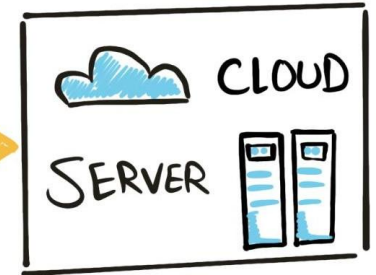
③
BUILD VIZ

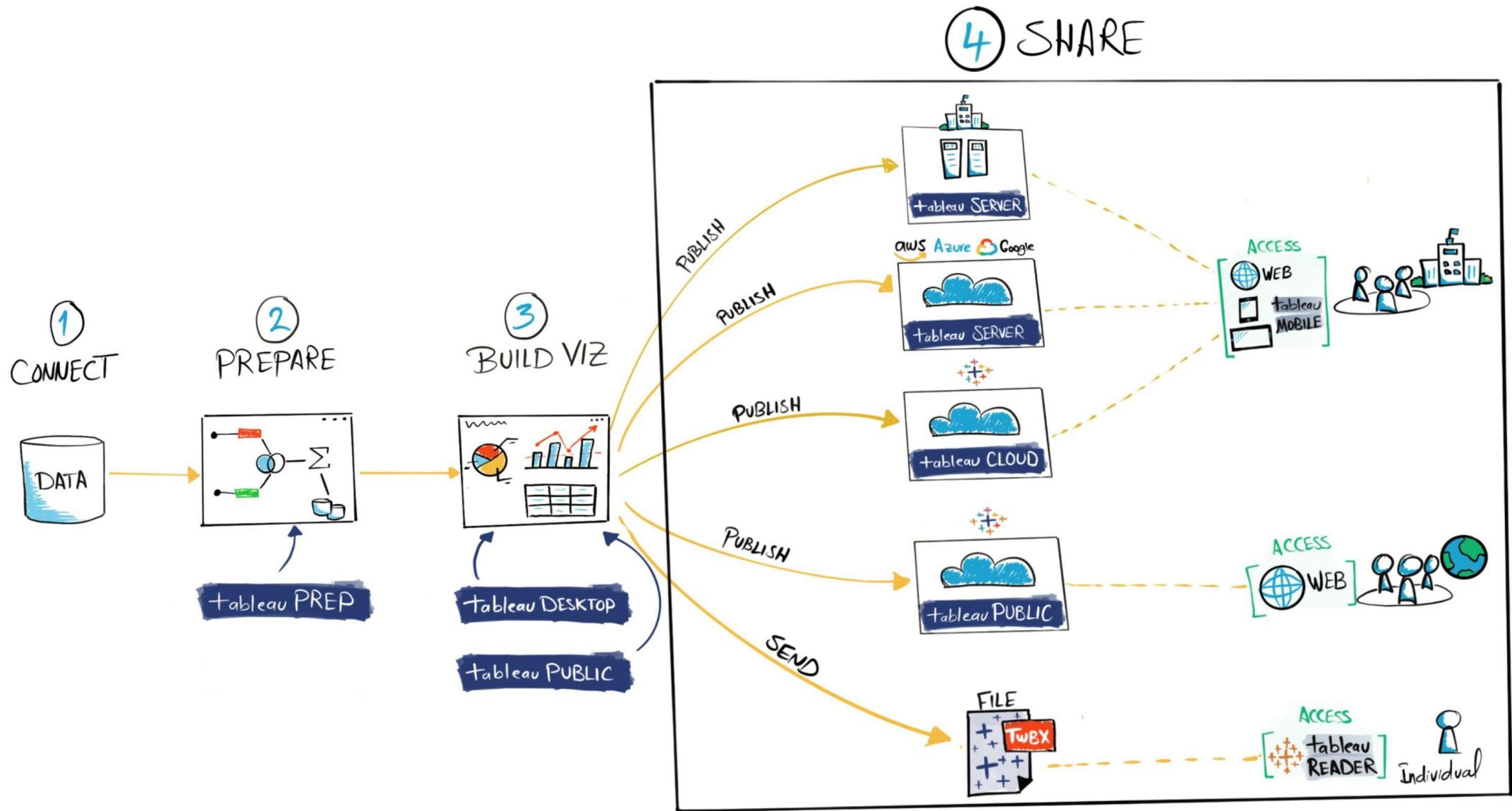


Tableau DESKTOP

Tableau PUBLIC

④
SHARE







DESKTOP

Tableau Desktop

"Backbone Product of Tableau"

Developer tool for data **visualization**

- **Connect** data
- **Build** views, dashboards, stories
- **Publish** workbooks & data sources

Requires a **license**

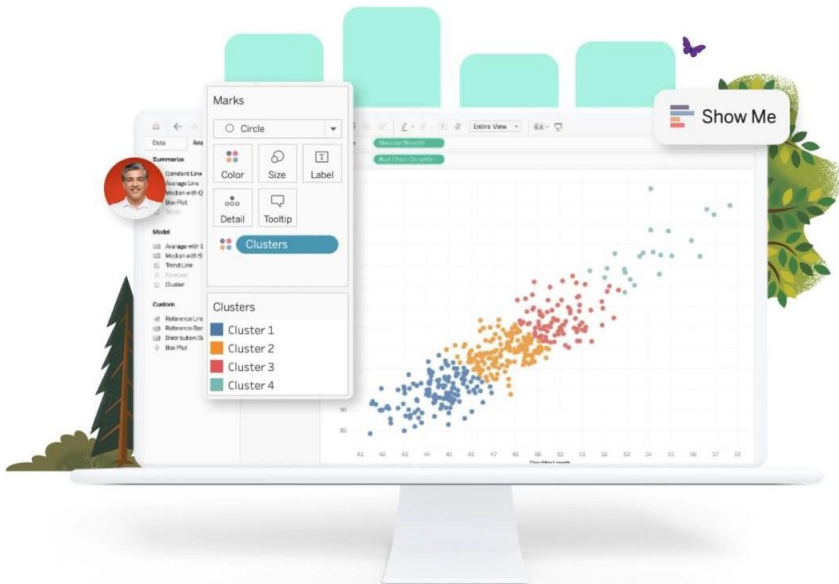
90 Data connectors

Publish

- Tableau **Server**
- Tableau **Cloud**
- Tableau **Public (cloud)**

No Limitations

For Data scientists & Data analysts





Developer tool for data visualization

- Connect data
- Build views, dashboards, stories
- Publish workbooks & data sources

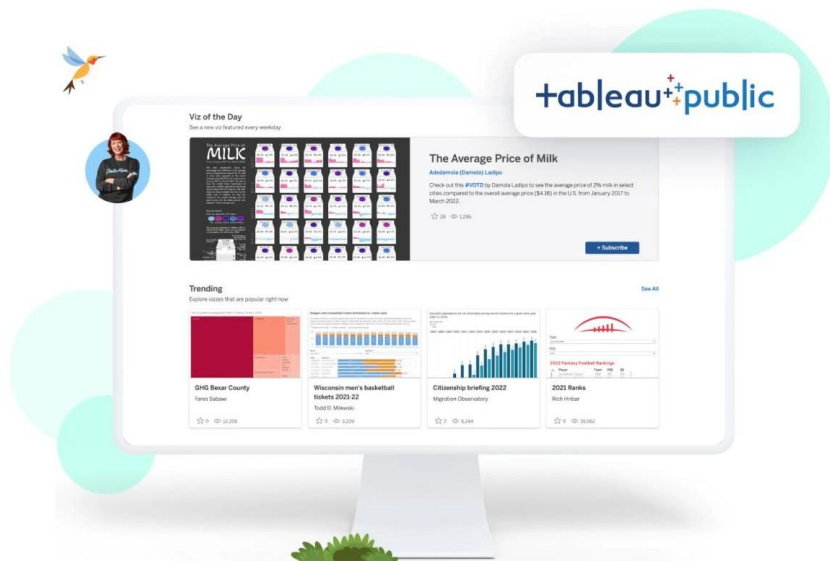
Free Tool

4 Main Limitations

- 10 Data connectors (only local files)
- Limited to 15 Million rows
- Publish to only Tableau Public (cloud)
- Can't save locally

All tools & functions to build Viz are available!

For Everyone



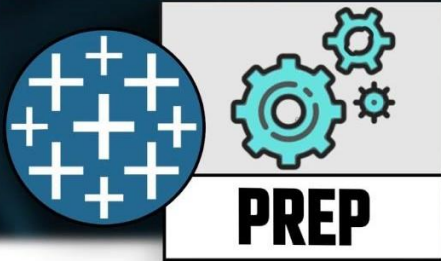


Tableau Prep

"Data Engineering Tool"

Developer tool for data engineering

- Connect data
- Build flows (clean, combine, aggregate ... etc)
- Publish flows & data sources

Requires a license

90 Data connectors

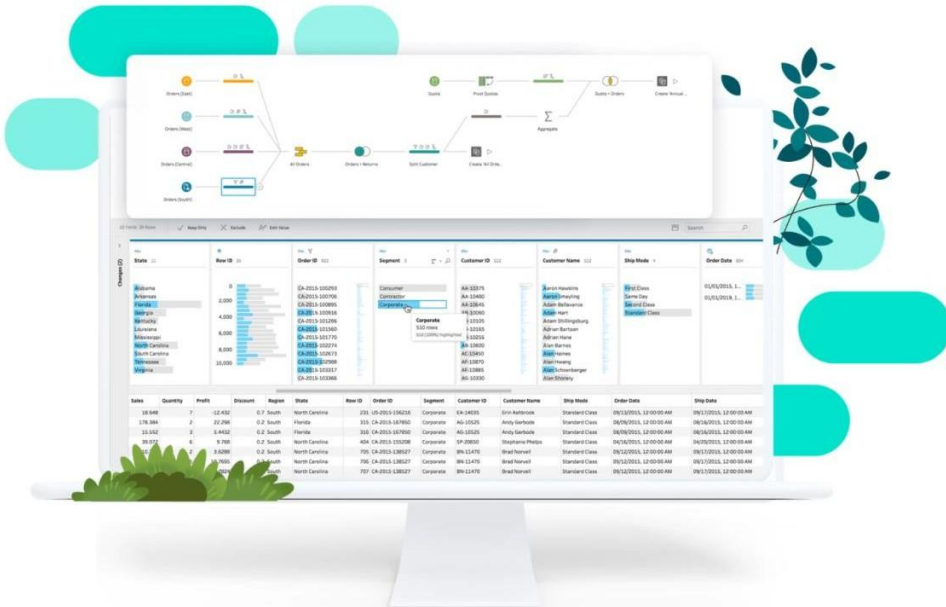
Output

- File
- Tableau Data Source
- Database Table

Publish

- Tableau Server
- Tableau Cloud

No public (free) version available.



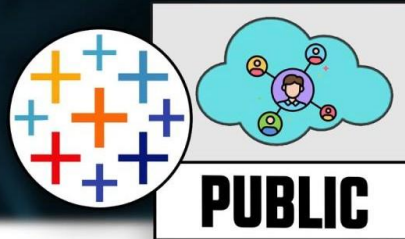


Tableau Public Cloud

"Social Media of Tableau"

Sharing tool for data visualization

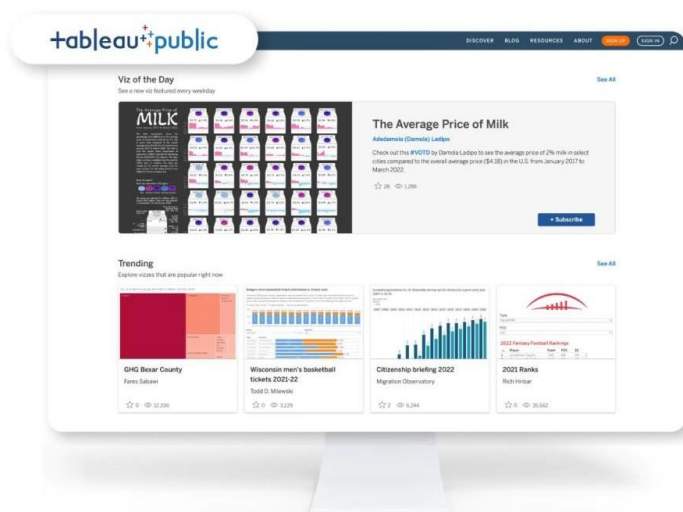
Free Cloud Service from Tableau

Platform

- Inspiring
- Connecting to other developers
- Sharing your skills

Main Limitations

- Limited to 10 GB for each account
- No connection to sources
- No Security features





READER

Tableau Reader

"Free Tool to view Vizzes"

Sharing tool for data visualization

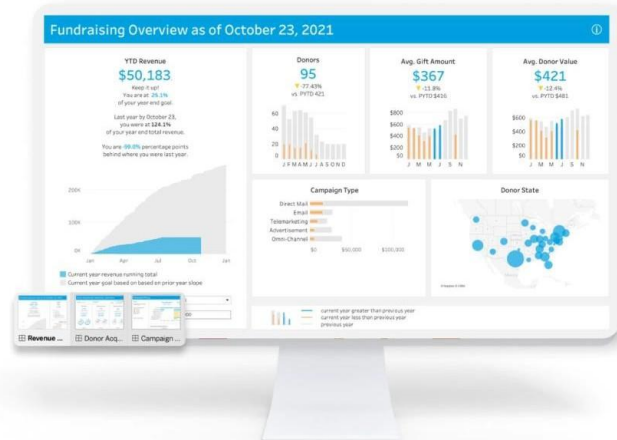
Free Tool

Only to view visualization

Main Limitations

- Can't build visualization
- Can't Refresh Data
- No Security features

Don't recommend using it in organization



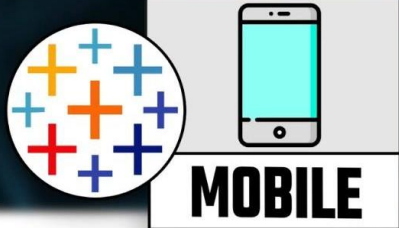
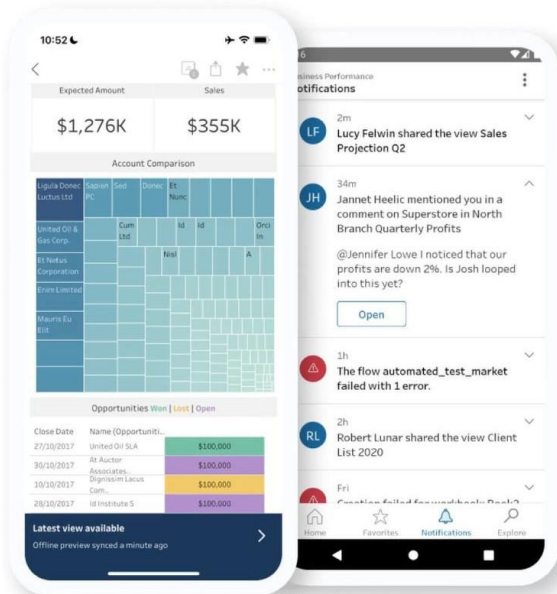


Tableau Mobile

"Access Vizs from anywhere"



Only to view visualization

Cant use it to build data visualization

Free App to download

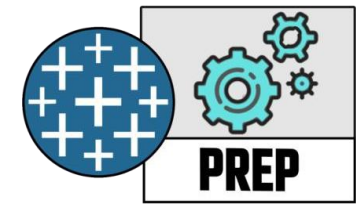
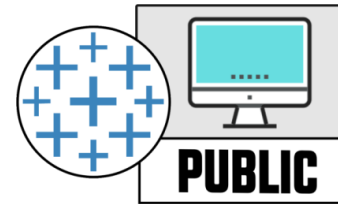
Require a licence to use

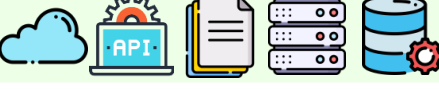
Connect to

- Tableau Server
- Tableau Cloud

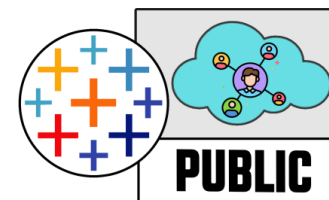
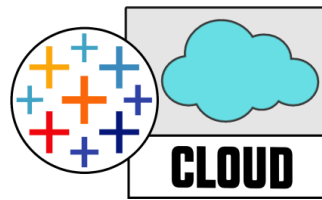
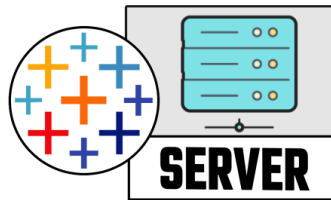
Caching Dashboards for offline access

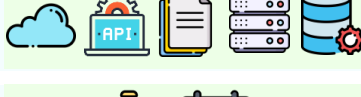
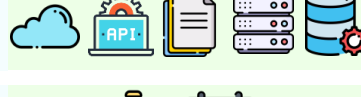
Dev Products Comparison



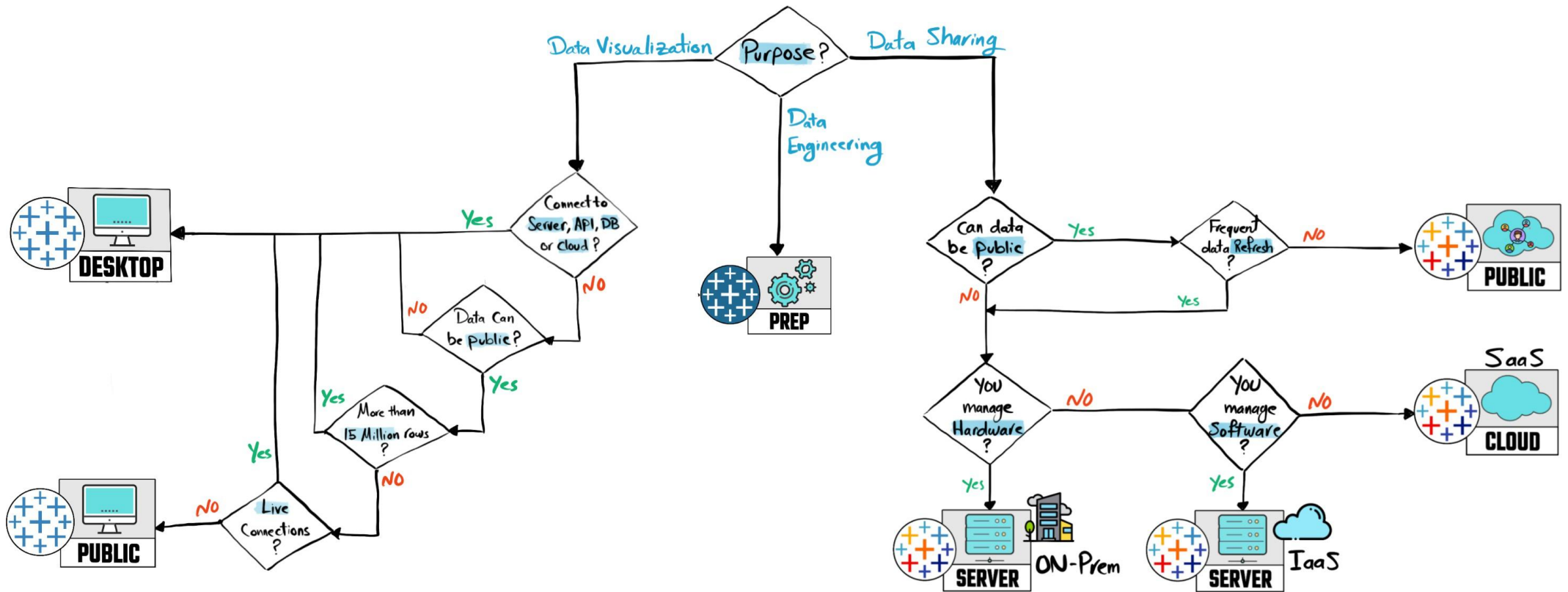
		DESKTOP	PUBLIC	PREP
 Purpose		Data Visualization	Data Visualization	Data Engineering
 Costs				
 Security				
 Data Limit		No Limit	15 Million Rows	No Limit
 Connectors				
 Live Con.				
 Store Locally				
 Audience		Data Scientists & Analysts	Everyone	Data Engineers

Sharing Products Comparison



	Project		 + a b e a u	 + a b e a u	
	Software				
	Hardware				
	Storage Limit	Server Disk Space	No Limits	10 GB	No Limits
	Connectors				
	Automation				
	Mobile				
	Sharing				

Choose Right Tableau Product



LIVE vs EXTRACT

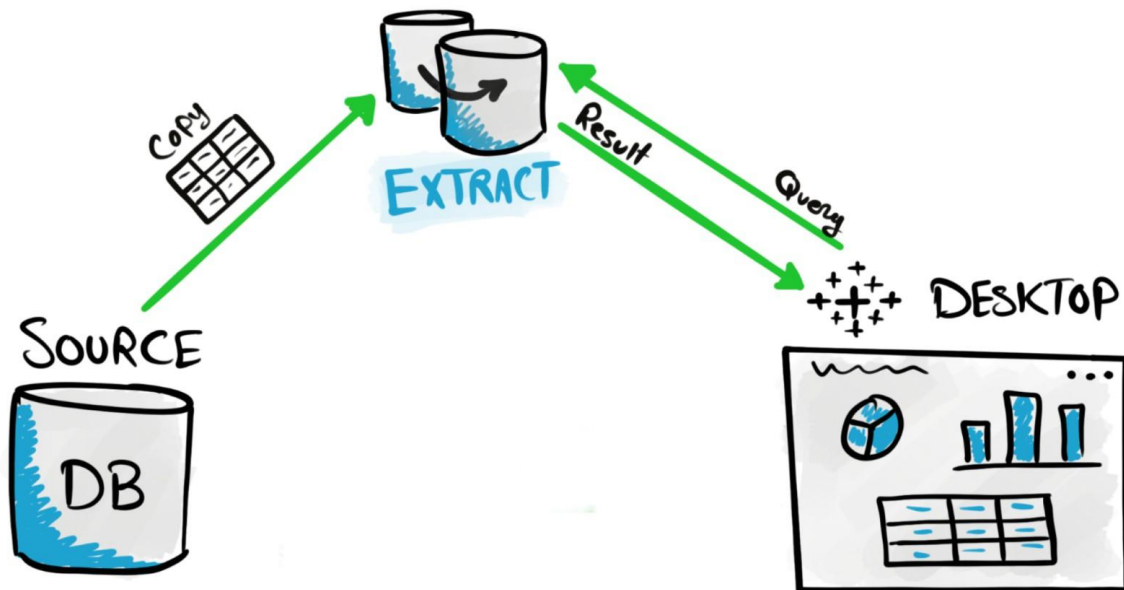
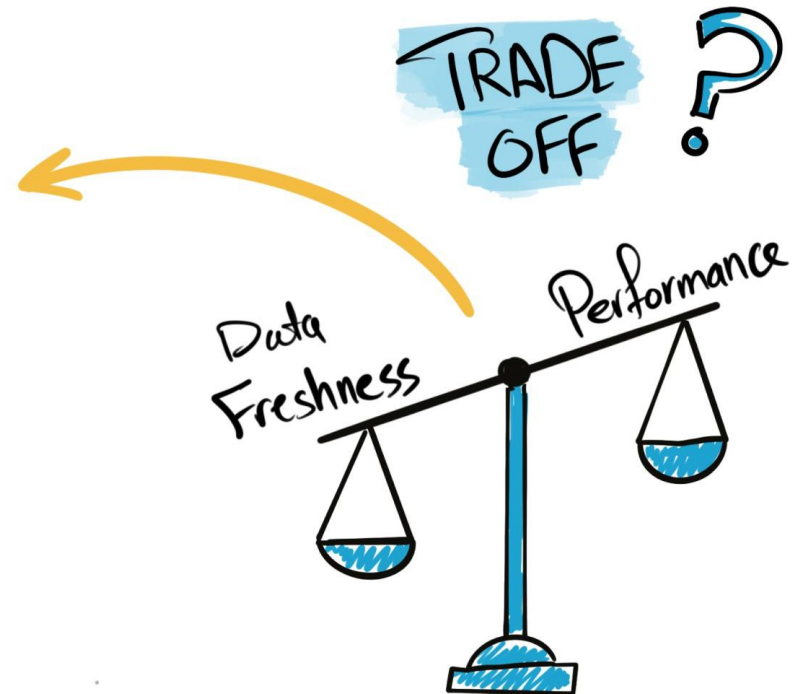
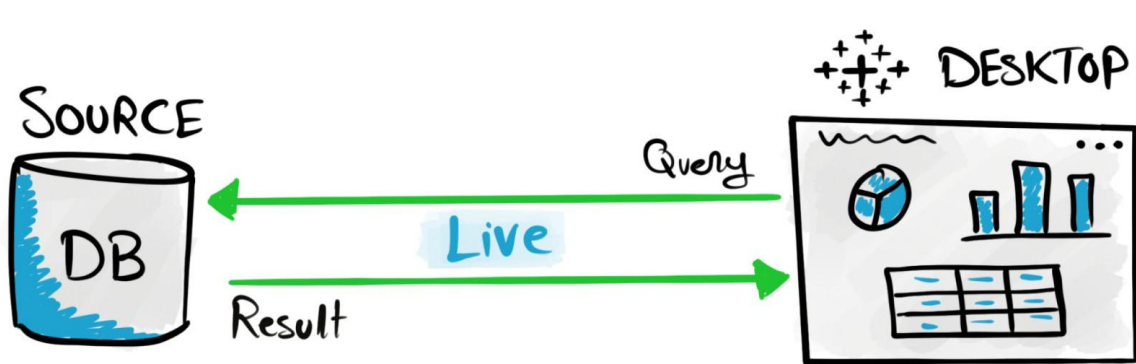


Tableau File Types

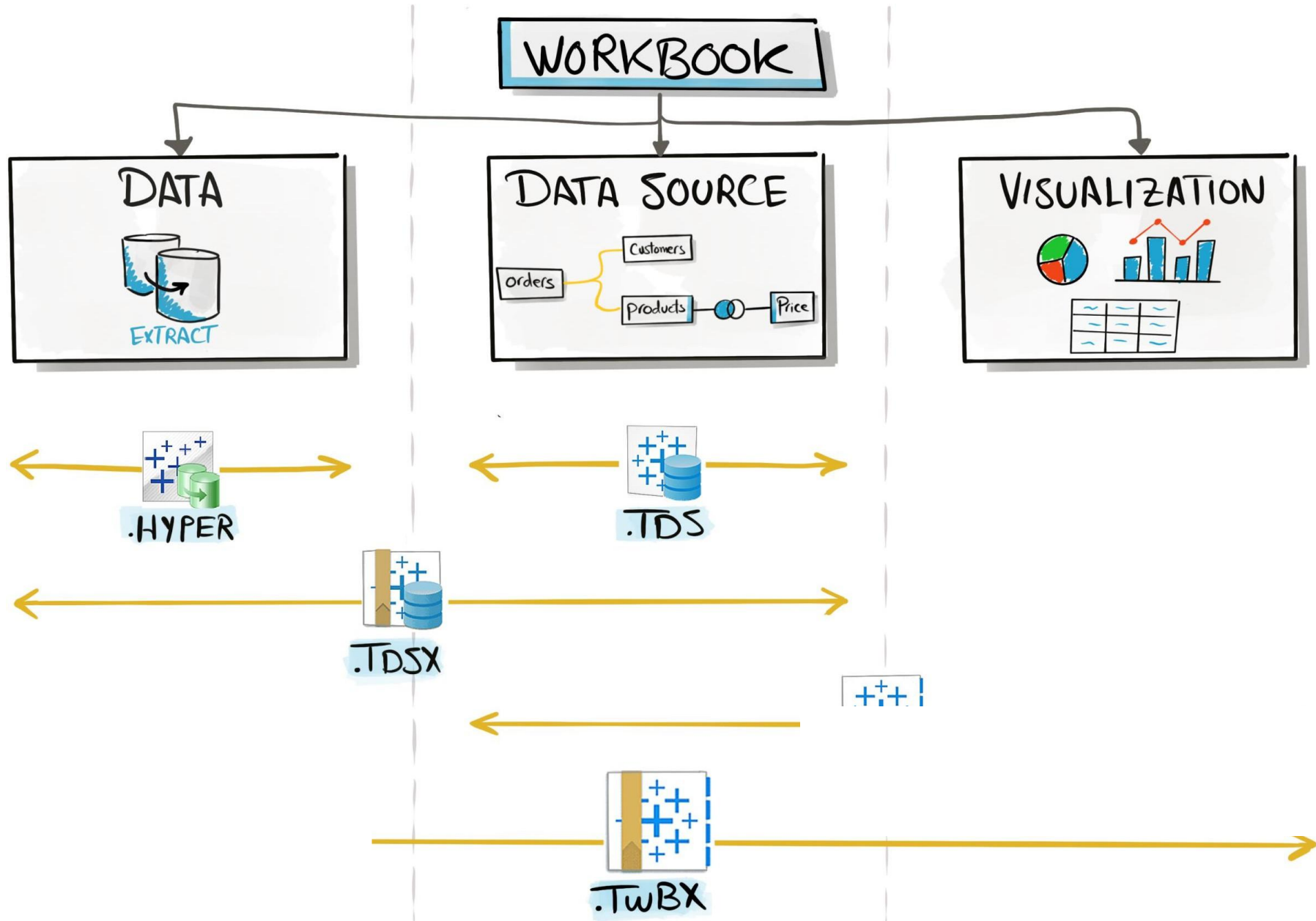
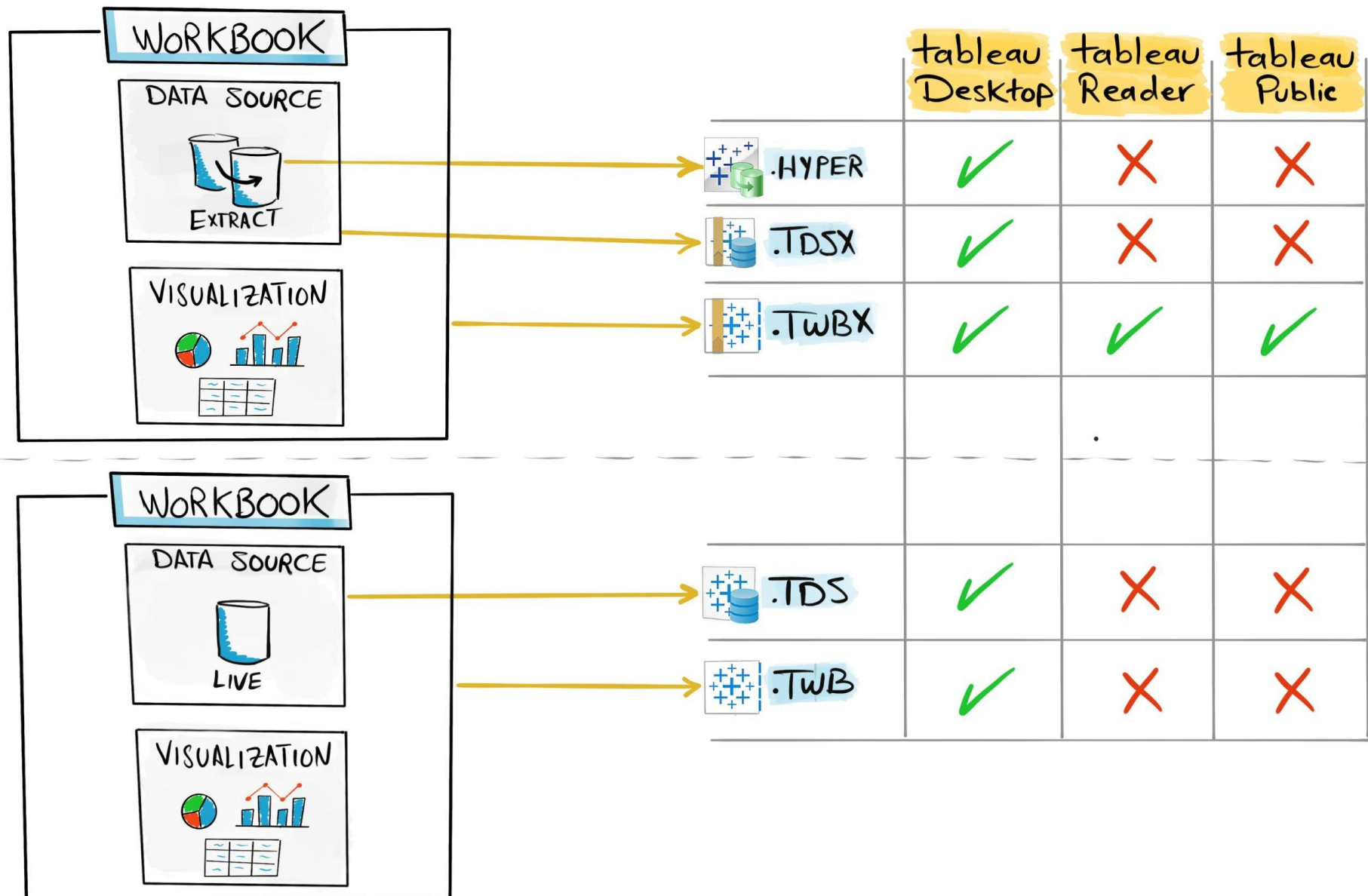
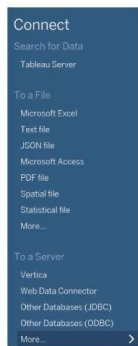


Tableau File Types & Products



DATA CONNECTORS



90 CONNECTORS

ACCESS INFO

E:\User\Data\Tableau\sales.csv

Path

DB Location

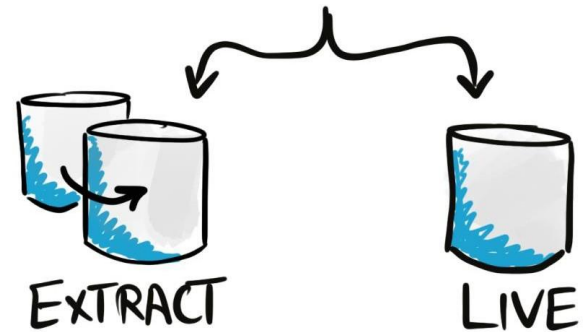
Server: localhost Port: 3306

Database: myfirstwebsitedb

Access token

czZCaGRSa3F0MzpnUjUjK09lqErXJW

CONNECTION TYPE



DATA MODEL

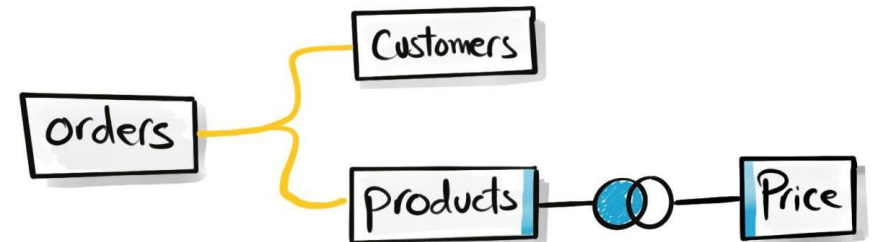


Tableau Workbook Files

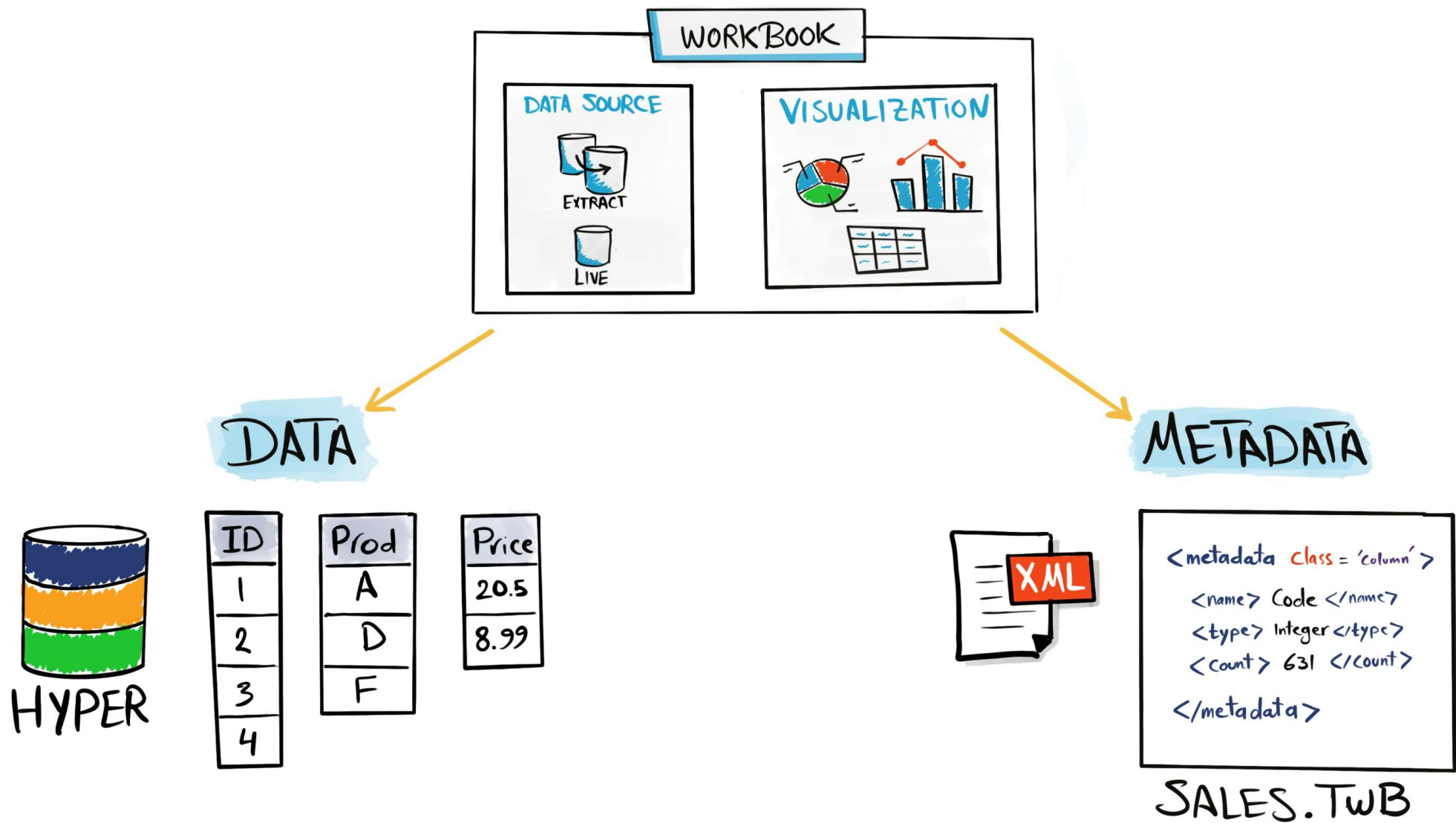


Tableau Architecture

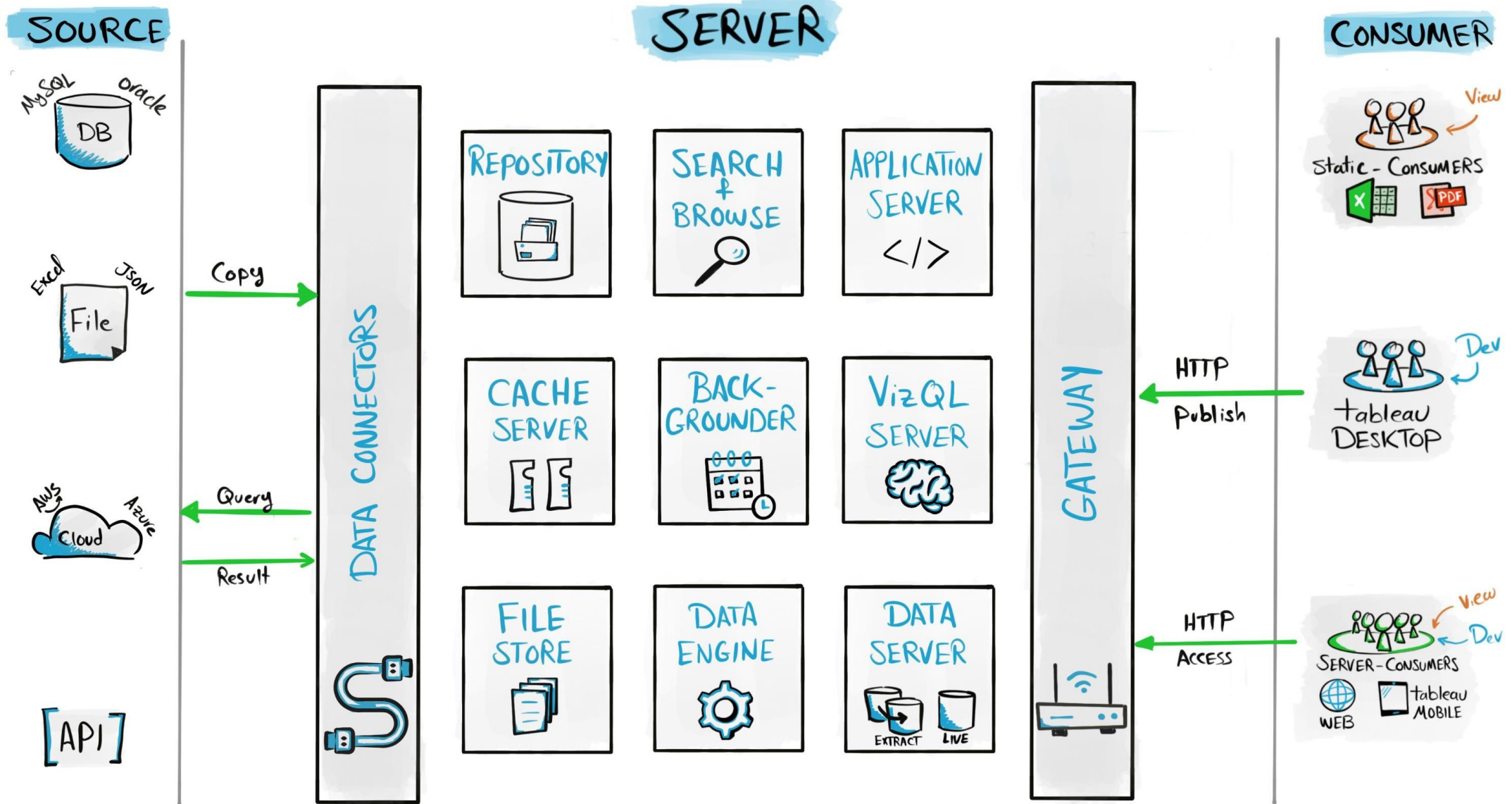
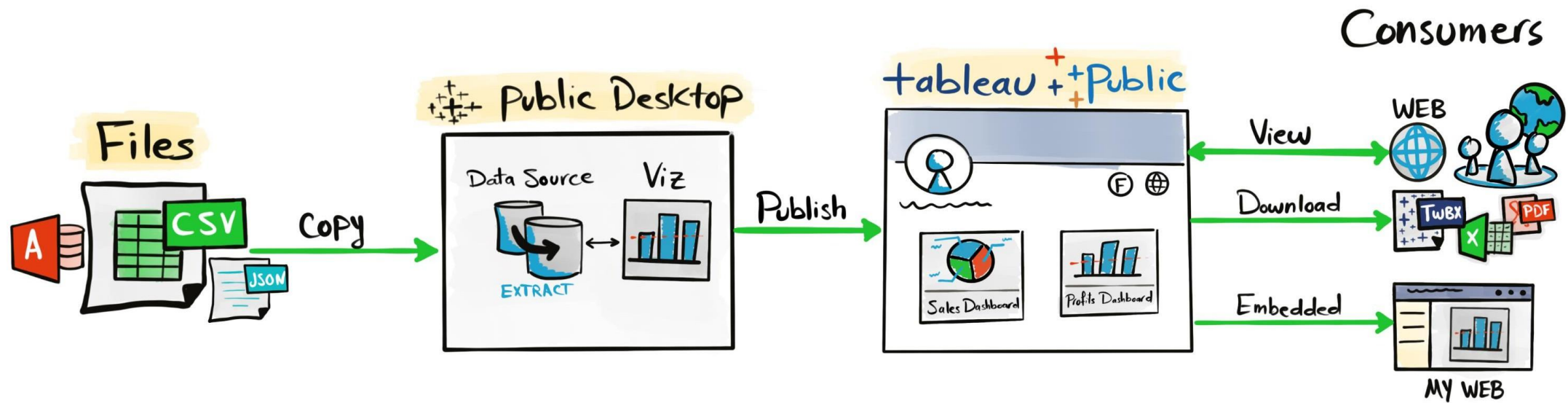
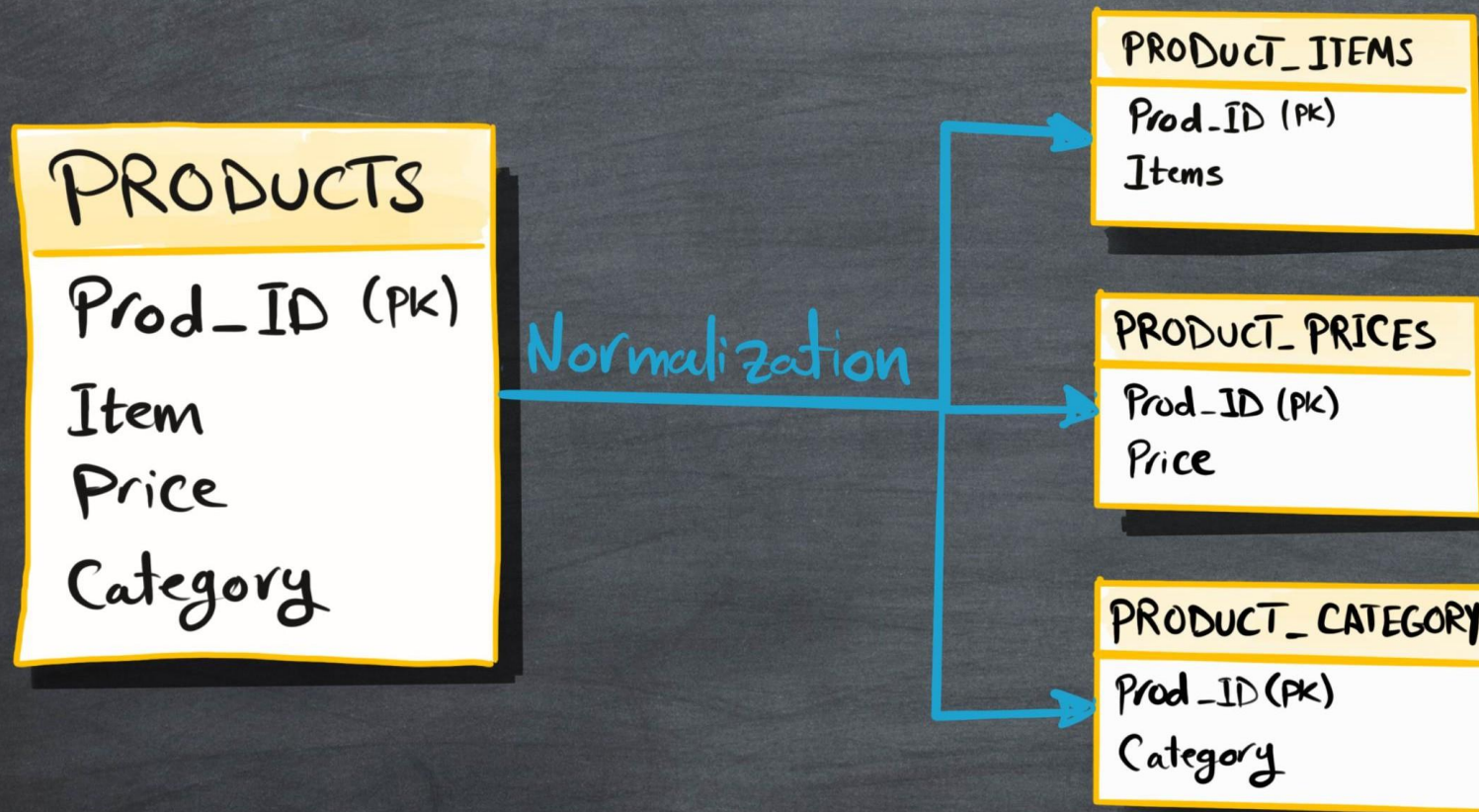


Tableau Public Architecture

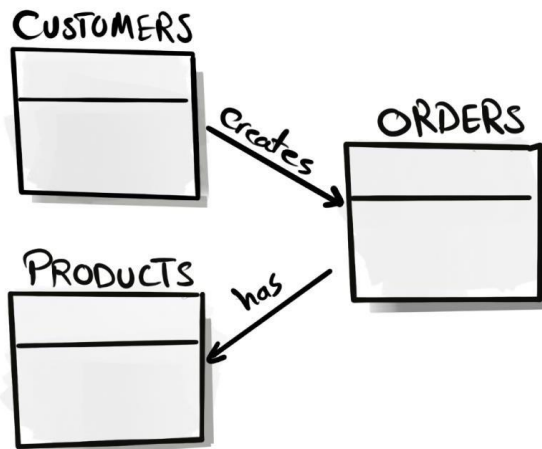


Data Normalization

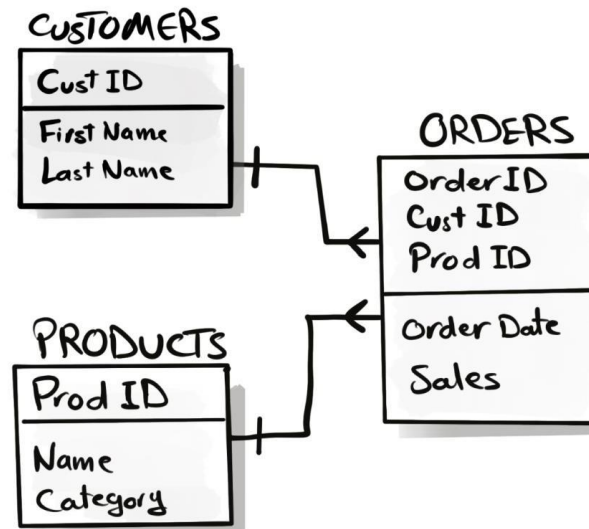


3 Types of Data Models

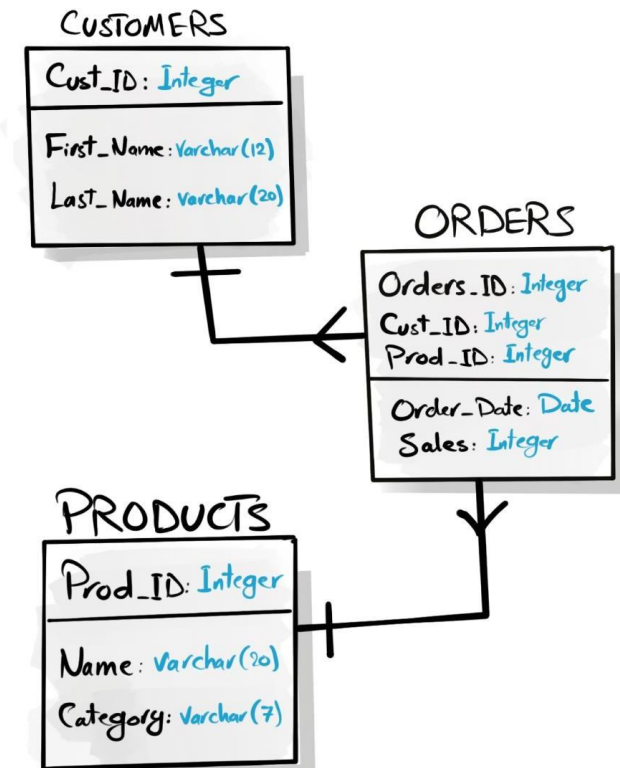
Conceptual Data Model



Logical Data Model

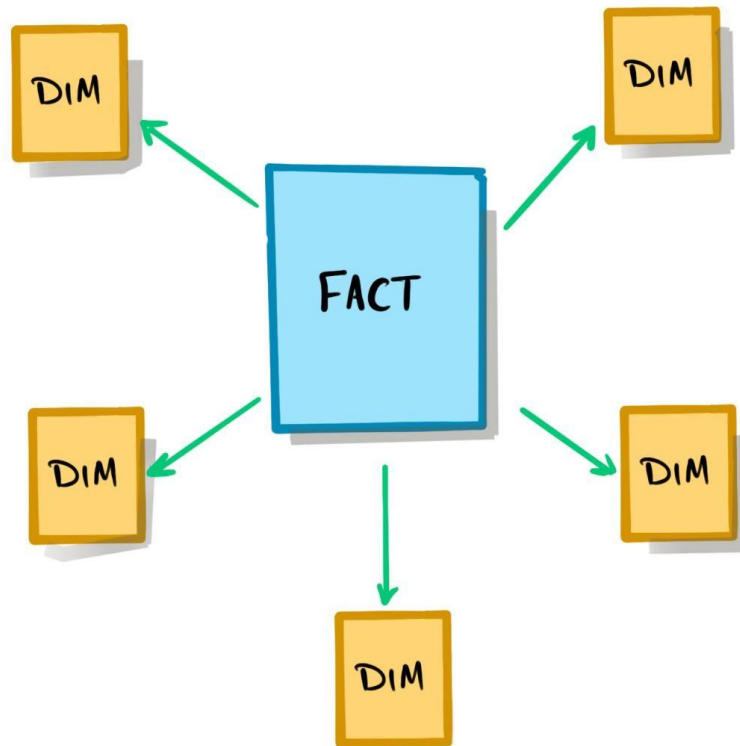


Physical Data Model

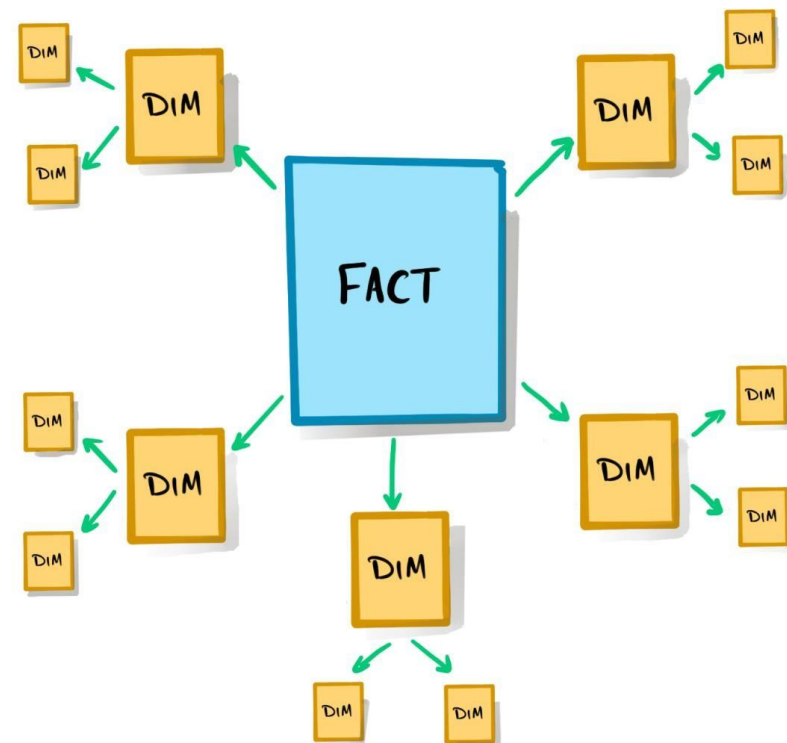


Multi-Dimensional Data Models

STAR SCHEMA



SNOWFLAKE SCHEMA



Data Model

STAR SCHEMA

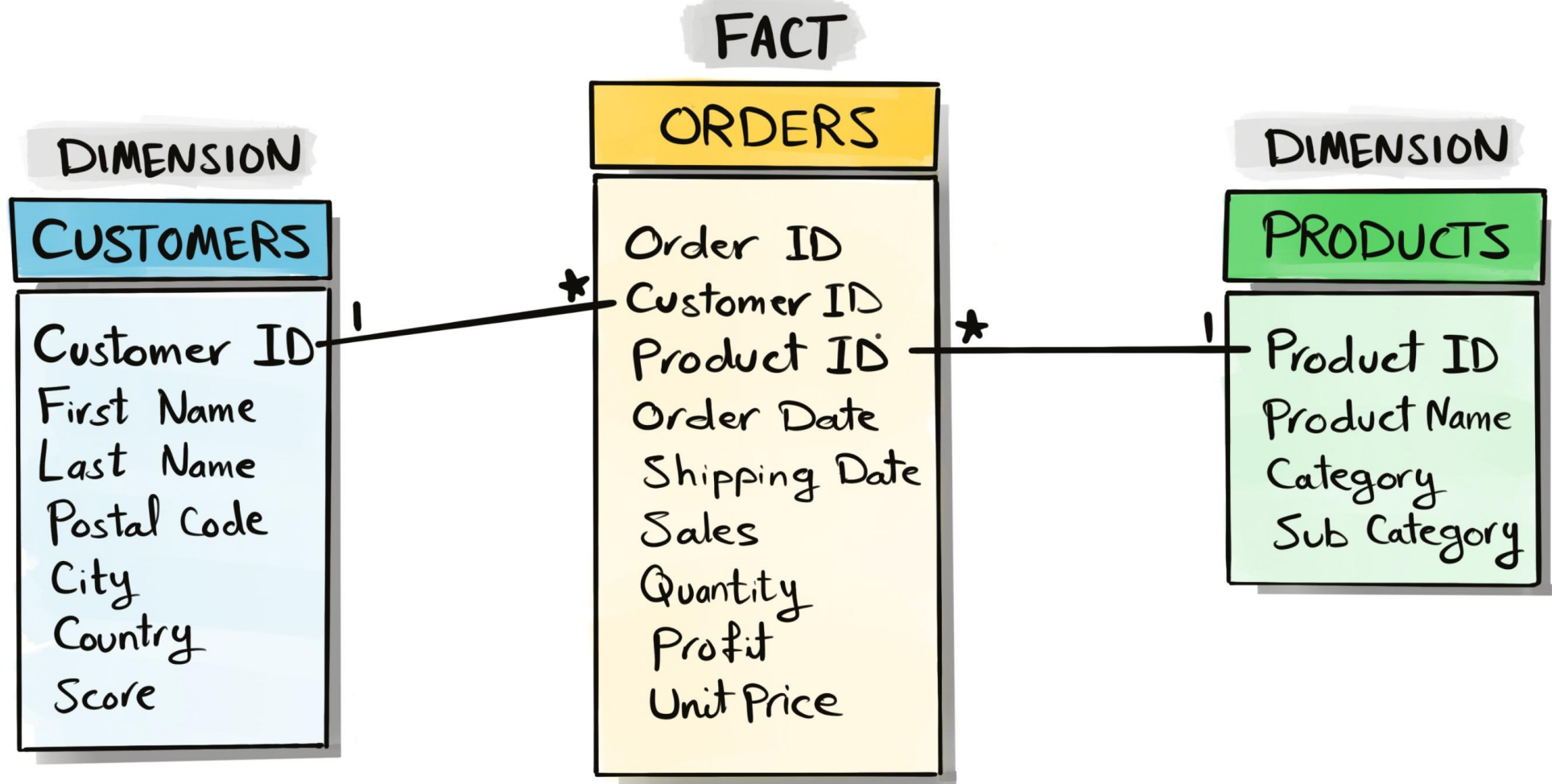
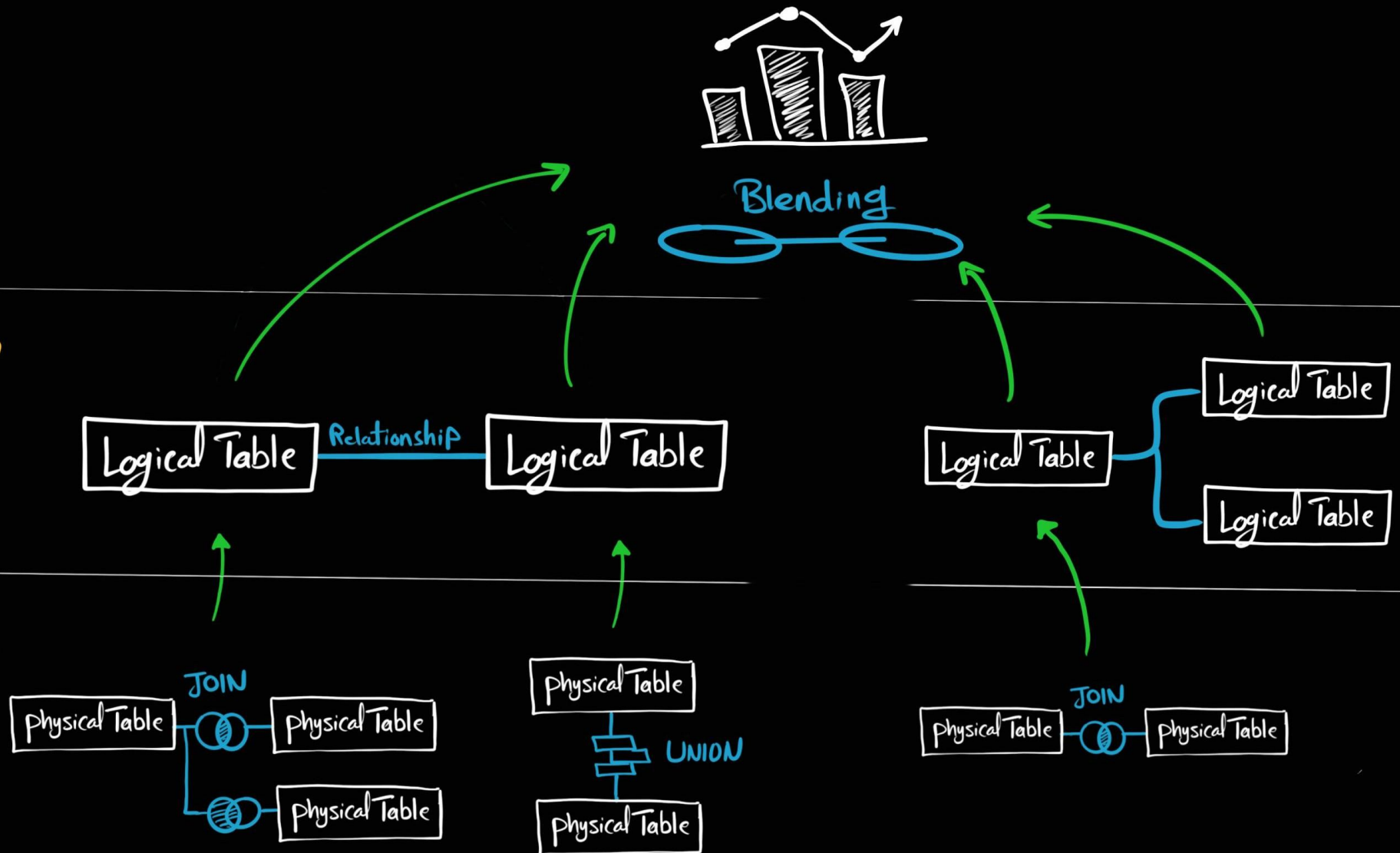


Tableau Data Models

VIZ

DATA SOURCE

Physical Layer Logical Layer

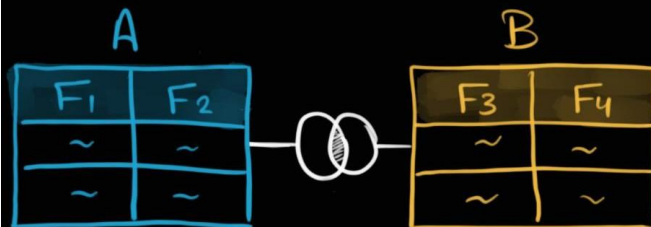


JOINS



DATA SOURCE

Physical Layer



NEW!

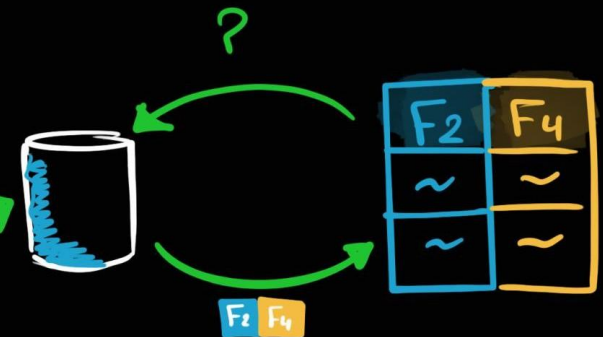
AB

F ₁	F ₂	F ₃	F ₄
~	~	~	~
~	~	~	~

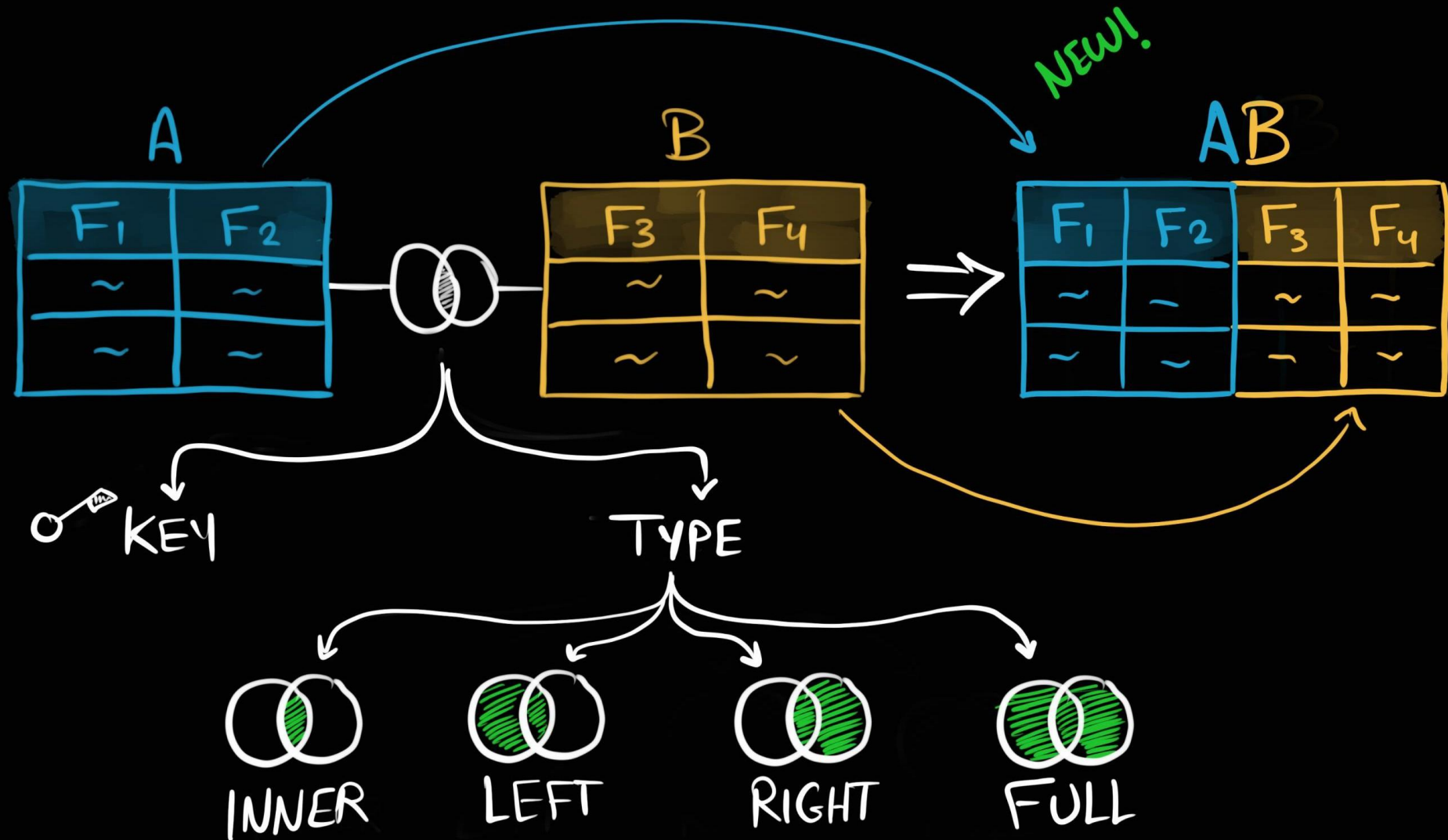
F₂ F₄



Visualization



JOINS



JOIN TYPES

Only Matching Rows ← A  B → All Rows

All Rows ← A  B → Only Matching Rows

Only Matching Rows ← A  B → Only Matching Rows

All Rows ← A  B → All Rows

JOIN | How it Works

CustomerNames

ID	Name
1	Maria
2	John

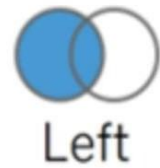


CustomerAges

ID	Age
1	20
3	35



ID	Name	ID	Age
1	Maria	1	20



ID	Name	ID	Age
1	Maria	1	20
2	John	NULL	NULL



ID	Name	ID	Age
1	Maria	1	20
NULL	NULL	3	35



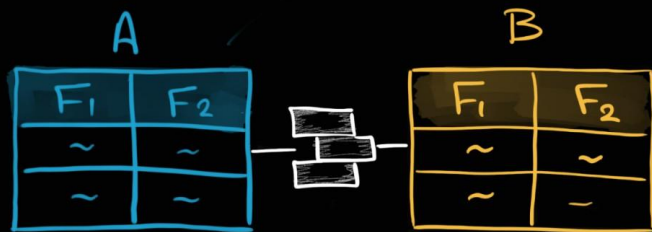
ID	Name	ID	Age
1	Maria	1	20
2	John	NULL	NULL
NULL	NULL	3	35

UNION



DATA SOURCE

Physical Layer



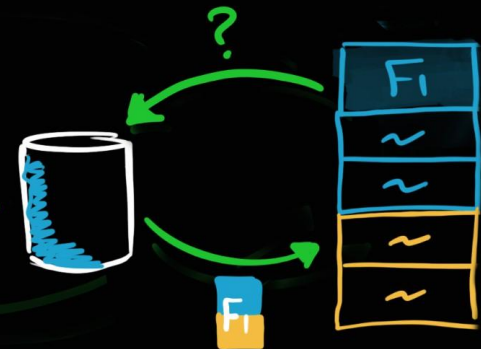
NEW!

AB

F ₁	F ₂
~	~
~	~
~	~

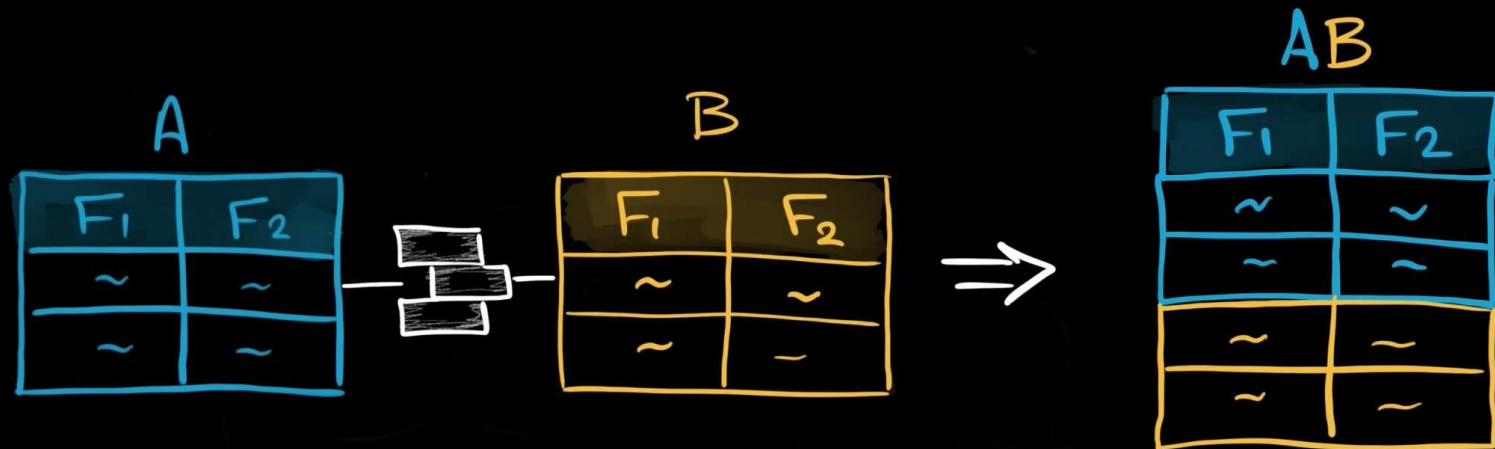
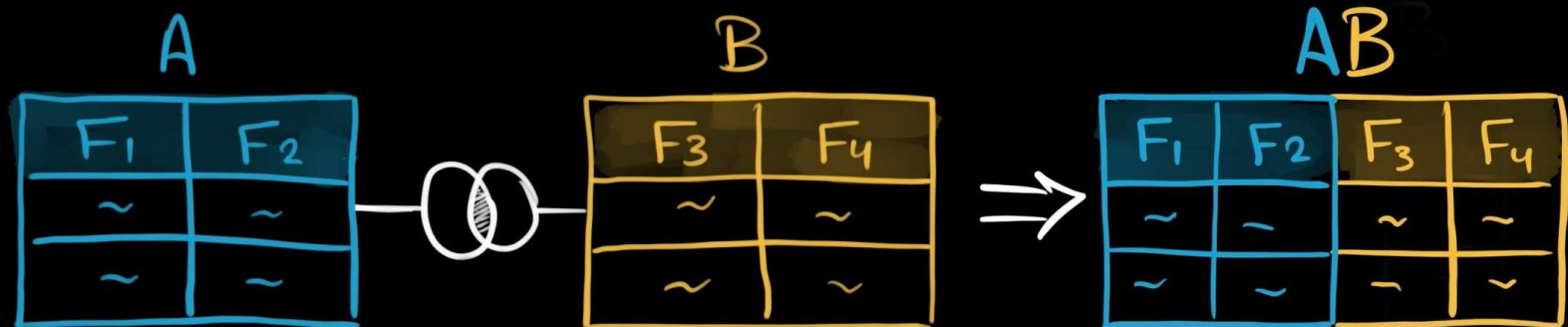


Visualization



JOINS vs UNION

JOIN Combine the FIELDS of two Tables



UNION Combine the ROWS of two Tables

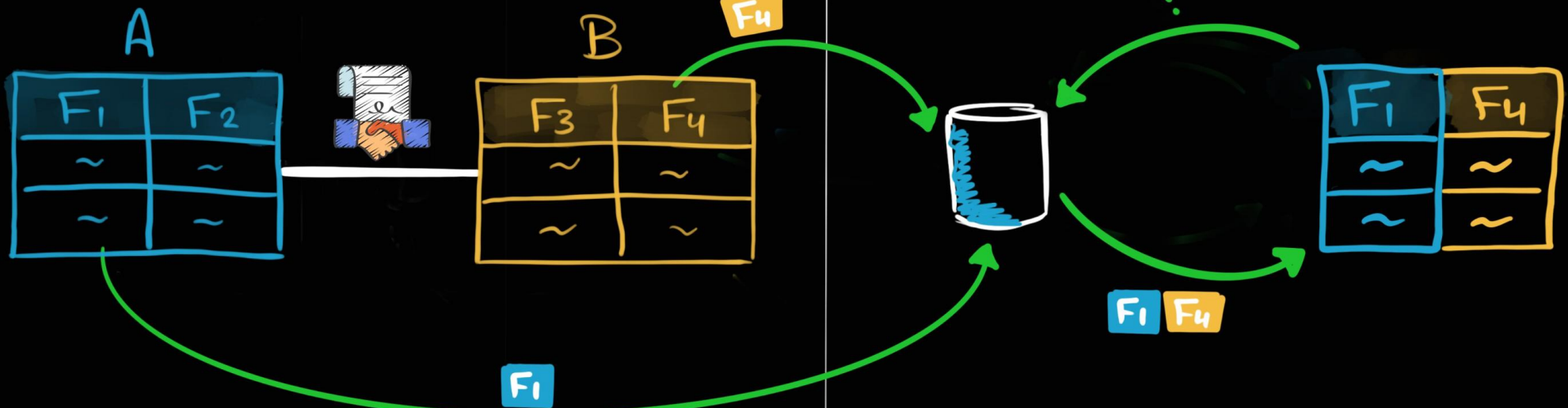
Relationships



DATA SOURCE
Logical Layer



Visualization



Relationship | Referential Integrity

LEFT TABLE	RIGHT TABLE	JOIN TYPES
Some Record Match	Some Record Match	
All Record Match	Some Record Match	
Some Record Match	All Record Match	
All Record Match	All Record Match	

Relationships | How it Works

Orders		Customers	
CustomerID	Sales	CustomerID	Name
1	50	1	Maria
2	20	2	John
2	10	3	Georg
1	30		

MANY-MANY	CustomerID	Sum (Sales)	Name
	1	80	Maria
	2	30	John

MANY-ONE	CustomerID	Sum (Sales)	Name
	1	80	Maria
	2	30	John

ONE-MANY	CustomerID	Sum (Sales)	Name
	1	50	Maria
	2	20	John

ONE-ONE	CustomerID	Sum (Sales)	Name
	1	50	Maria
	2	20	John

Correctness Performance

OK	NOK
----	-----

OK	OK
----	----

NOK	NOK
-----	-----

NOK	OK
-----	----

Cardinaltiy

MANY	MANY
MANY	ONE
ONE	MANY
ONE	ONE

Data Blending



DATA SOURCE
Logical Layer

A

F ₁	F ₂
~	~
~	~

B

F ₃	F ₄
~	~
~	~

F₁

Primary



F₁

LEFT JOIN

F ₁	F ₄
~	~
~	~

F₄

Secondary



F₄

Joins vs Blending



JOINS

First **Combine** Then **Aggregate**



BLENDING

First **Aggregate** Then **Combine**

Customers

CustomerID	Name
1	Maria
2	John
3	Georg



Left

Orders

CustomerID	Sales	Date
1	50	2020
1	30	2023
2	20	2023



CustomerID	Name	CustomerID	Sales	Date
1	Maria	1	50	2020
1	Maria	1	30	2023
2	John	2	20	2023
3	Georg	NULL	NULL	NULL

Customers

CustomerID	Name
1	Maria
2	John
3	Georg



BLENDING

Orders

CustomerID	Sales	Date
1	50	2020
1	30	2023
2	20	2023



CustomerID	Name	CustomerID	Sales	Date
1	Maria	1	80	*
2	John	2	20	2023
3	Georg	NULL	NULL	NULL

Joins vs Relationships



JOINS

STATIC

DATA LOSS



RELATIONSHIP

FLEXIBLE

NO DATA LOSS

Joins vs Relationships

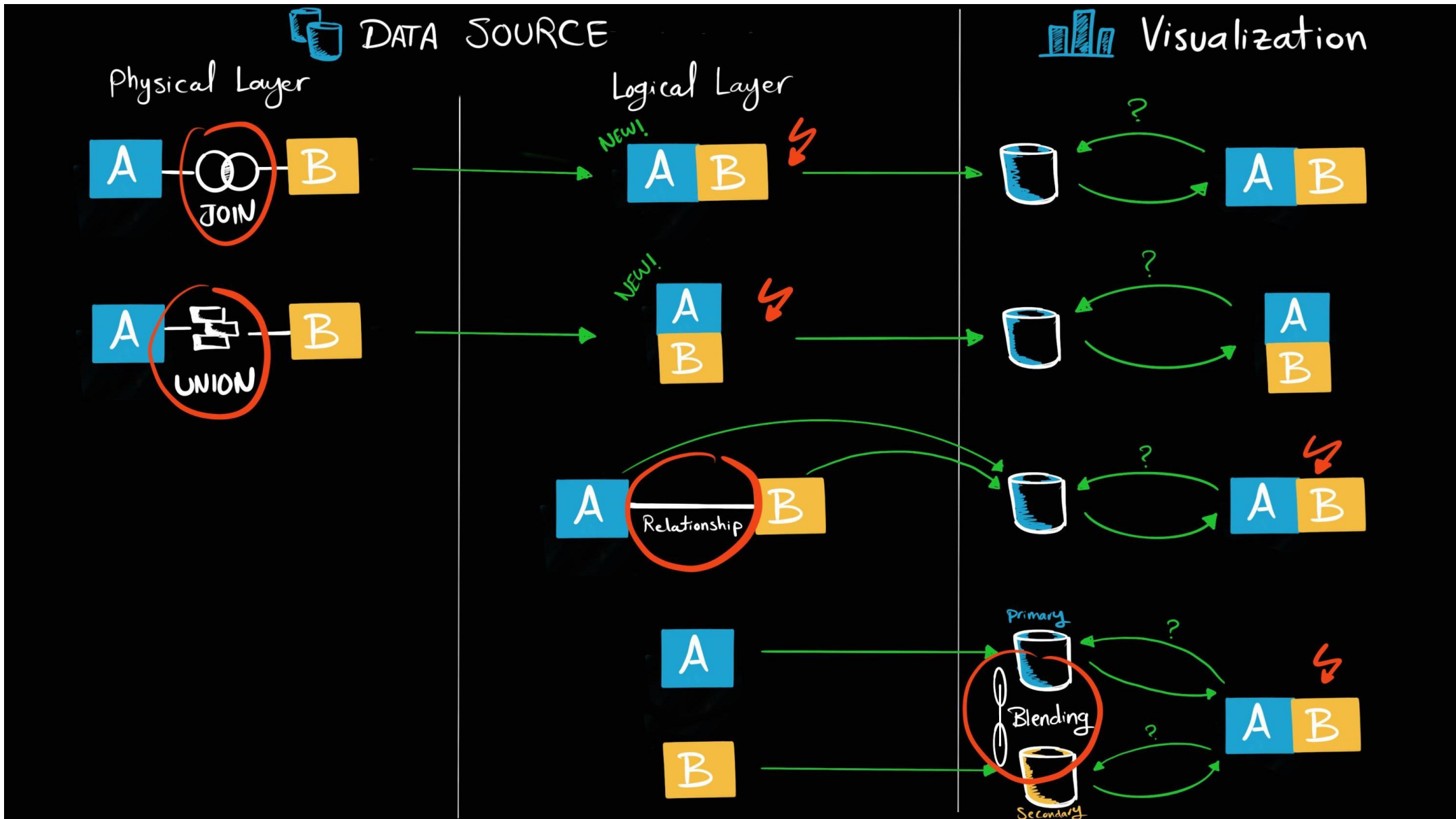
THE PROBLEM

Merging Tables may cause **Duplicates**

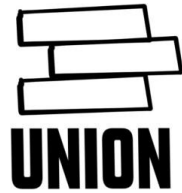
&

Duplicates cause **false Aggregations**

Summary | Combine Methods



Summary | Combine Methods



Layer

Physical Layer

Physical Layer

Logical Layer

Visualization Level



Multi-Data Sources



Merge Tables?



Flexibility

Static

Static

Flexible

Very Flexible
(Page by Page)



Join Types

n/a

Inner, Left, Right, Full

All

Left



Rank

n/a

3

1
(Default)

2

Put it all together

Let us create a Data
Source on top of the
small data set